FinSight

Relational Schema Diagram

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Implemented schema: 17 application tables + bookkeeping + risk view + Mongo telemetry

app_user

user_id	email	password_hash	display_name	role	is_active
last_login_at	created_at				

customer

customer_id	user_id	full_name	date_of_birth	email	phone
country_code	status	kyc_verified_on	updated_at		

FK user_id ⟶ app_user.user_id

address

address_id	customer_id	line1	city	state	postal_code
country_code	latitude	longitude	is_primary		

FK customer_id ⟶ customer.customer_id

account

account_id	customer_id	account_number	account_type
currency_code	balance	status	opened_at	updated_at

FK customer_id ⟶ customer.customer_id

payment_method

method_id	account_id	method_type	token	last4	expiry_date
status	updated_at				

FK account_id ⟶ account.account_id

app_session

session_id	user_id	created_at	last_seen_at	idle_expires_at
absolute_expires_at	revoked_at	revocation_reason		

FK user_id ⟶ app_user.user_id

app_session_cleanup

operation_id	admin_id	preview_at	cutoff_at	expires_at
candidate_session_ids	candidate_count	status	idempotency_key	
request_hash	completed_at	deleted_count		

FK admin_id ⟶ app_user.user_id

schema_migrations

version	filename	checksum	applied_at

merchant

merchant_id	merchant_name	category_code	country_code
city	latitude	longitude	risk_tier	is_watchlisted	updated_at

device_fingerprint

device_id	device_hash	os_family	first_seen_country
first_seen_at			

transaction

transaction_id	account_id	method_id	merchant_id	device_id
amount	currency_code	exchange_rate	amount_base	direction
txn_type	status	occurred_at	merchant_country_code	
merchant_city	latitude	longitude	description	search_vector

FK (account_id,currency_code) ⟶ account(account_id,currency_code)
FK device_id ⟶ device_fingerprint.device_id
FK merchant_id ⟶ merchant.merchant_id
FK (method_id,account_id) ⟶ payment_method(method_id,account_id)

risk_rule

rule_id	rule_code	rule_name	dimension	weight
threshold_value	is_active	version		

risk_signal

signal_id	transaction_id	rule_id	observed_value
score_contribution	evidence	triggered_at	

FK rule_id ⟶ risk_rule.rule_id transaction_id ⟶ transaction.transaction_id

fraud_scenario

scenario_id	scenario_code	scenario_name	description

transaction_label

transaction_id	scenario_id	is_suspicious

FK scenario_id ⟶ fraud_scenario.scenario_id transaction_id ⟶ transaction.transaction_id

investigation_case

case_id	case_ref	transaction_id	assigned_to_user_id	status
priority	resolution_code	resolution_notes	opened_at	due_at	
closed_at	updated_at	version			

FK assigned_to_user_id ⟶ app_user.user_id transaction_id ⟶ transaction.transaction_id

case_event

event_id	case_id	actor_user_id	event_type	from_state	to_state
reason	metadata	operation_id	event_ordinal	occurred_at	

FK actor_user_id ⟶ app_user.user_id case_id ⟶ investigation_case.case_id

case_command_receipt

operation_id	actor_user_id	idempotency_key	command_type
request_hash	case_id	response_status	completed_at

FK actor_user_id ⟶ app_user.user_id case_id ⟶ investigation_case.case_id

v_transaction_risk

transaction_id	risk_score	signal_count	risk_band

device_telemetry [Mongo]

_id	device_id	transaction_id	captured_at	schema_version
shape_type	retention_eligible	payload		

connection_telemetry : payload

ip_routing_hash	connection_type	latency_ms	tls_cipher
vpn_detected			

client_environment : payload

screen_resolution	battery_level	battery_charging	os_patch_level
hardware_concurrency	user_agent_category		

KEY: solid underline PK; dotted underline FK.
Red arrows stay inside their own relation.
FK source -> target is written below each relation.
Mongo references are not SQL FKs.
Bookkeeping/view are implementation supplements.

All attributes retained. FK continuations explicitly name both source and target. No cross-page lines or shared buses. Cream cells are generated.